

Machine Learning-Based Design of Load-Independent WPT Systems

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Abstract—This paper proposes a machine learning (ML)–based design for a load-independent (LI) wireless power transfer (WPT) system. The fundamental concept of the proposed design strategy is to solve an optimization problem for achieving high-frequency LI operation by full numerical computations. An evaluation function for optimization is formulated, taking into account output voltage regulation against load variations and high power-delivery efficiency. The developed optimization software provides the best parameter set of coupling coil parameters and circuit-component values. In addition, the ML-based optimization finds the optimal parameter set from parameters outside the range that it had considered to be common sense, which is an unexpected but valuable result. The WPT system, designed through the proposed method, achieved ideal LI operation in the experiments. The quantitative agreements between experimental and numerical waveforms substantiate the validity of the proposed design.

Index Terms—Machine learning, load-independent, wireless power transfer

I. INTRODUCTION

High-frequency wireless power transfer (WPT) systems in the megahertz band have been developed, which has an advantage for miniaturizing the system sizes [1]. Switching losses increase proportionally in high-frequency operation, and zero-voltage switching (ZVS) is critical to achieving high efficiency [2]. Besides, output-voltage regulation against load variations is required for practical applications. Namely, in high-frequency WPT systems, achieving output-voltage regulation and high-efficiency operation simultaneously against load variations is a significant research challenge. Typically, a feedback control is applied to address this issue [3]. However, output-voltage information feedback from the receiver to the transmitter via wireless communication introduces propagation delays and packet losses, degrading the stability and reliability of the control. Besides, it is complex and difficult for the output regulation and the ZVS to be achieved simultaneously through such a type of feedback control. It is the ideal form that the output-voltage regulation and ZVS can be achieved without feedback control.

Recently, the load-independent (LI) technology [4]–[9] has attracted many researchers' attention. The LI technology can solve the above issues from the roots, namely the LI WPT system achieves a constant output voltage and the ZVS against load variations without any control. To realize the LI operation, we have to obtain a proper parameter set, which is obtained from analytical expressions [8], [9].

In general, the analytical expressions should be based on several assumptions for the formulation of the analytical model. For example, all the analyses of the LI WPT system assume pure sinusoidal resonant current. When the WPT operates outside of these assumptions, differences appear between the theoretical predictions and the experimental results. The small output voltage variation occurs against load variations in the actual circuit due to the parasitic components. In addition, experimental power-delivery efficiency deteriorates compared to the theoretical prediction. To obtain accurate performance predictions in the frontloading design step, the numerical design approach proved to be effective [10]. This is because the numerical design does not require any assumptions to simplify the analysis. There is, however, no numerical design example for a LI WPT system until now.

This paper proposes a machine learning (ML)–based numerical design strategy for a LI WPT system. The basic concept of the proposed design is that the WPT design is reduced to an optimization problem, which is solved by fully numerical computations. An evaluation function for the LI WPT is formulated, which expresses the output-voltage regulation and high-efficiency operation. The evaluation function is minimized by adjusting the design parameters, namely the physical parameters of the magnetic components and the component values of the WPT circuit. Because it is possible to include effects of the parasitic component effect and low loaded quality factor in the system design, the WPT system, designed through the proposed method, achieved a consistent output voltage and high power-delivery efficiency compared with the analysis-based designed system. In addition, highly accurate performance prediction is possible due to the numerical design approach, which is validated by the quantitative agreements between experimental results and numerical predictions. The ML-based optimization finds the optimal parameter set from parameters outside the range that it had considered to be common sense, which is an unexpected but valuable result.

II. WPT SYSTEM USING LI INVERTER

There has been extensive research on the LI operation [4]–[9]. Fig.1(a) shows the circuit diagram of the WPT system, which is consisted of the class-EF inverter, the S-S coupling, and the class-D rectifier.

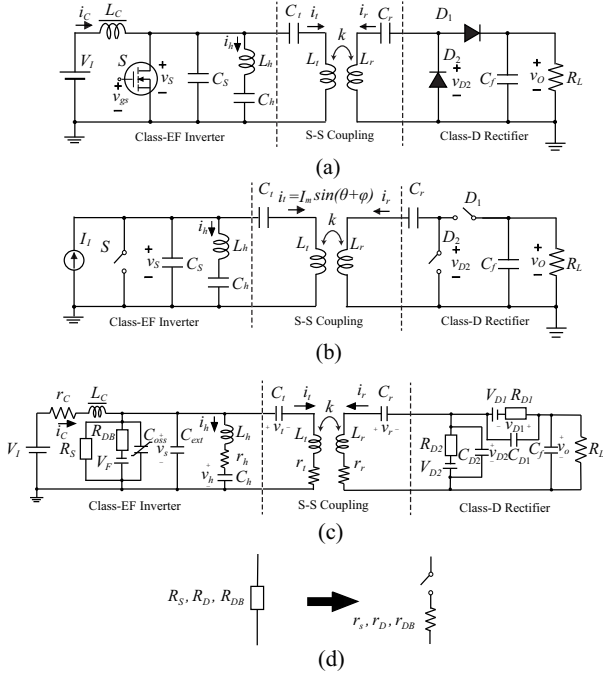


Fig. 1. WPT system with the class-EF inverter. (a) Circuit topology. (b) Analytical model. (c) Numerical model. (d) Modeling of the switching device.

Fig.2 shows the example waveforms of a WPT system, where $\theta = \omega t = 2\pi ft$ is the angular time, ω is the transmission angular frequency, and f is the transmission frequency. D_S is the off duty ratio, and R_{Lr} is the resistance at the rated condition. The LI class-EF inverter achieves ZVS and constant amplitude of the current flowing through the transmission coil regardless of the load resistance, as shown in Fig.1(b). As a result, a constant output voltage is obtained regardless of load.

LI operation is achieved through analysis under specific conditions. It ensures that the ZVS condition is met and the inverter output current remains unaffected by load variations. This analysis is based on an ideal circuit model with several key assumptions, as illustrated in Fig. 1(b). For instance, it presumes a pure sinusoidal resonant current and disregards parasitic components. However, in practical circuits, the operation deviates from the ideal state due to the influence of parasitic components in the devices and elements. Additionally, the generation of a pure sinusoidal resonant current does not occur. As a result, the output voltage varies depending on load variations. In addition, the analytical design does not optimize power-delivery efficiency for the parasitic resistance of the passive devices, resulting in lower power-delivery efficiency at light loads. Therefore, it is necessary to establish a design methodology for LI WPT systems independent of assumptions.

III. ML-BASED DESIGN

A. Circuit Modeling

Fig. 1(c) shows an equivalent circuit model of a WPT system considered parasitic components: whole the Equivalent

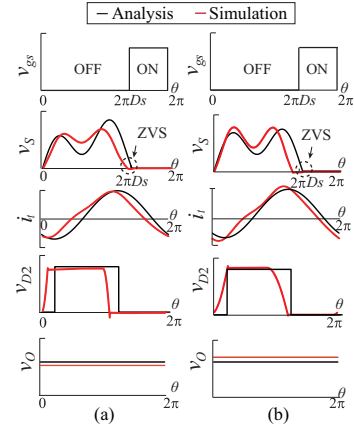


Fig. 2. Example of operating waveforms of WPT system with LI EF-class inverter. (a) $R_L/R_{Lr} = 1$. (b) $R_L/R_{Lr} = 10$.

Series Resistance (ESR) of the inductor, the on-resistance and the parasitic capacitance of switch and diode, and forward voltage drops of diode.

The switch model R_S varies with the angular time and is defined as follows,

$$R_S = \begin{cases} \infty, & \text{for } 0 \leq \theta \leq 2\pi D, \\ r_{on}, & \text{otherwise,} \end{cases} \quad (1)$$

where r_{on} is the on-resistance of the switch. The diode models R_{Dj} and R_{DB} vary with each on-resistance or infinite according to the diode voltage. The parasitic capacitance C_{oss} of the switch S is modeled as the variable capacitance depending on the switch voltage.

The coupling coils L_t and L_r are modeled as air gap solenoid coupling coils. The self-inductance and ESR can be derived from physical parameters of coils; such as number of turns N , coil diameter d , and air gap g [11] [12], and Neumann's formula can numerically derive the mutual inductance [13]. The inductor L_h is modeled as a toroidal coil with a core [14]. The inductance and ESR can be derived by using such as AL-value A_L and number of turns N .

B. Overview of the Proposed Design

Fig. 3 shows a flowchart of the proposed design. The fundamental concept of the proposed design strategy is to solve an optimization problem for achieving high-frequency LI operation by full numerical computations.

For designing the WPT system, the state variable vector is

$$\mathbf{x} = [i_C, v_S, i_h, v_h, v_t, i_t, i_r, v_r, v_{D1}, v_o]^T, \quad (2)$$

and circuit parameters vector is

$$\boldsymbol{\lambda} = [\omega_t^*, \omega_r^*, \omega_2^*, \gamma_s, N_t, N_r, N_h, w_t, w_r, w_h, Core]^T, \quad (3)$$

where definitions of vector elements are coil parameters and circuit-component values. Then, the circuit equations in Fig. 1(c) can be expressed as

$$\frac{d\mathbf{x}}{d\theta} = \mathbf{f}(\theta, \mathbf{x}, \boldsymbol{\lambda}). \quad (4)$$

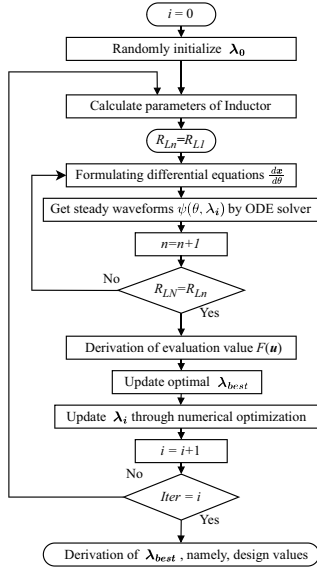


Fig. 3. Flowchart of the proposed design.

By using a numerical integration algorithm in a period $0 \leq \theta \leq 2\pi$, the system waveforms can be derived. If the integral interval is large enough, waveforms converge into the steady state where $\psi(\theta, \lambda)$ is defined as the steady-state system waveforms. A vector, which is a candidate of the solution for optimal design, is defined as

$$\mathbf{u} = [\psi(\theta, \lambda)^T, \lambda^T]^T, \quad (5)$$

The design problem is to find the proper vector $\mathbf{u}$ for maximizing the evaluation function, which is described in the next subsection. The proper vector searching is carried out by ML. Namely, all the proposed design processes are performed by fully numerical computations.

C. Power-Delivery Efficiency

The input power P_I and output power P_O are derived by the following equations.

$$P_I = \frac{V_I}{2\pi} \int_0^{2\pi} i_C(\theta) d\theta \quad (6)$$

$$P_O = \frac{1}{2\pi R_L} \int_0^{2\pi} v_o^2(\theta) d\theta \quad (7)$$

From (6)-(7) and core loss P_{core} of L_h in [15], the power-delivery efficiency η can be estimated by the following equation.

$$\eta(\mathbf{u}) = \frac{P_O}{P_I + P_{core}} \quad (8)$$

D. Evaluation Function

The purpose of the circuit design is to derive circuit parameters that achieve constant output voltage and high power-delivery efficiency against load variations $R_{L1} \leq R_L \leq R_{L2}$ without using assumptions as in analysis-based design. To

TABLE I
DESIGN SPECIFICATIONS OF WPT SYSTEM FOR EXPERIMENT.

Symbol	Value	Coil	Value
Input voltage V_I	80 V	Diameter L_t	50 mm
Operating frequency f	6.78 MHz	Diameter L_r	46 mm
Reference of output voltage V_{Or}	16.8 V	Height L_t	15 mm
Minimum load resistance R_{L1}	14.7 Ω	Height L_r	15 mm
Off-duty ratio D_S	0.7	Gap	15 mm

achieve the above purpose, we propose the evaluation function as

$$\min_{\mathbf{u}} F(\mathbf{u}) = \alpha F_\eta(\mathbf{u}) + (1 - \alpha) F_V(\mathbf{u}), \quad (9)$$

Subject to $N_t w_t < h_t$ and $N_r w_r < h_r$,

where $F_\eta(\mathbf{u})$ and $F_V(\mathbf{u})$ are evaluation functions of power-delivery efficiencies and the output voltages, and α is a weighting coefficient. Besides, $F_\eta(\mathbf{u})$ and $F_V(\mathbf{u})$ are definitions as

$$F_\eta(\mathbf{u}) = 1 - \frac{\eta_1(\mathbf{u}) + \eta_2(\mathbf{u})}{2}, \quad (10)$$

$$F_V(\mathbf{u}) = \frac{F_{V1}(\mathbf{u}) + F_{V2}(\mathbf{u})}{2},$$

where $\eta_1(\mathbf{u})$ and $\eta_2(\mathbf{u})$ are power-delivery efficiencies at $R_L = R_{L1}$ and $R_L = R_{L2}$, respectively. Additionally, the output voltage functions $F_{V1}(\mathbf{u})$ and $F_{V2}(\mathbf{u})$ are defined as

$$F_{V1}(\mathbf{u}) = \left| 1 - \frac{V_{O1}(\mathbf{u})}{V_{Or}} \right|, \quad (11)$$

$$F_{V2}(\mathbf{u}) = \left| 1 - \frac{V_{O2}(\mathbf{u})}{V_{O1}(\mathbf{u})} \right|,$$

where $V_{O1}(\mathbf{u})$ and $V_{O2}(\mathbf{u})$ are average of output voltages at $R_L = R_{L1}$ and $R_L = R_{L2}$, respectively. V_{Or} is the reference of the output voltage.

In circuit design, maintaining precise output voltage regulation is often more critical than maximizing power-delivery efficiency, as even a slight deviation in output voltage can lead to more significant performance issues than a similar reduction in power-delivery efficiency. In this paper, we set $\alpha = 0.2$ to prioritize output voltage accuracy over efficiency.

The proposed evaluation function does not directly express the achievement of ZVS, which differs from the analytical design constraints that guarantee ZVS. However, switching loss increases in high-frequency operation. It is assumed that maximizing power-delivery efficiency implicitly means achieving the ZVS. The circuit parameters that the proposed evaluation function maximized can be expected to have performance equal to or better than the analysis-based design.

IV. CIRCUIT EXPERIMENT

Table I gives the design specifications for the experimental verifications in this paper. The ML-based design provides optimized circuit parameters, as given in Table II. Table II also gives the circuit parameters obtained from the previous analysis-based design [9]. The proposed design optimizes coils and circuit parameters simultaneously.

Fig. 4 shows the experimental waveforms of the LI WPT systems. As we can see from Fig. 4 (a) and (c), i_t was not a

TABLE II
DESIGN PARAMETERS.

Param.	Definition	Analytical	Proposed
ω_t^*	$1/(\omega\sqrt{L_t C_t})$	0.66	0.13
ω_r^*	$1/(\omega\sqrt{L_r C_r})$	1.04	0.97
ω_h^*	$1/(\omega\sqrt{L_h C_h})$	1.78	0.50
γ_s	$\omega C_S R_{L1}$	0.12	0.11
N_t	Turn number of L_t	5	6
N_r	Turn number of L_r	3	4
N_h	Turn number of L_h	27	32
w_t	Wire diameter of L_t	0.8 mm	1.8 mm
w_r	Wire diameter of L_r	1.0 mm	1.8 mm
w_h	Wire diameter of L_h	1.0 mm	0.75 mm
Core	Core of L_h	-	T-106-6

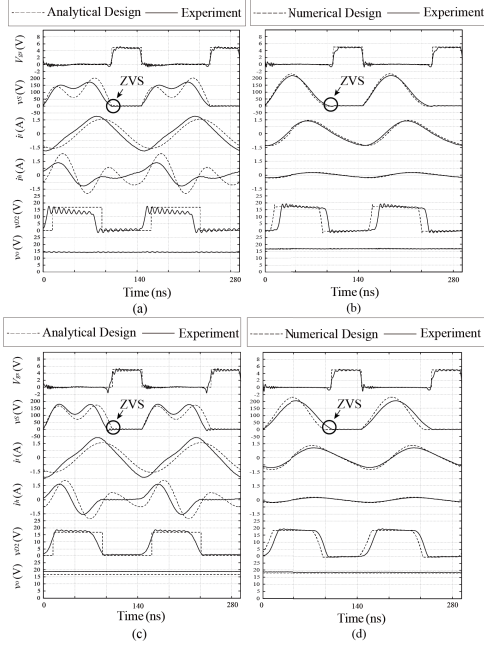


Fig. 4. WPT waveforms. (a) and (c) Analysis-based design. (b) and (d) Proposed ML-based design. (a) and (b) For $R_L/R_{L1} = 1$. (c) and (d) For $R_L/R_{L1} = 10$.

pure sinusoidal wave. Additionally, a transition of the diode voltage v_{D2} is affected by parasitic capacitors. These indicate differences between the experiment and the analytical model. Although the amplitude i_t of the inverter current based on the analytical design is known to be constant regardless of load variations [9], it was confirmed that the amplitude i_t increases and the output voltage increases accordingly. On the other hand, Fig. 4(b) and (d) show that the switch voltage v_s achieved the ZVS despite the non-including achievement of ZVS in the proposed evaluation function. The output voltage v_o was constant regardless of the load resistance in the WPT system designed by the proposed design strategy. The numerical design method contributes to the high-performance prediction accuracy, and the quantitative agreement between the experimental and numerical prediction results verifies its validity. These results indicate that the proposed design could search for the parameters that achieve the load-independent amplitude of the current i_t in the experiments. It can be

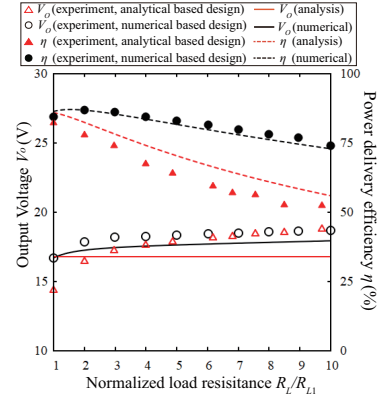


Fig. 5. Output voltage and power-delivery efficiency as functions of normalized load resistance.

confirmed that the current flowing in L_h is significantly reduced in Fig. 4(b) and (d) compared to Fig. 4 (a) and (c). It is considered the result of optimization to reduce the loss due to parasitic resistance in L_h without harmonic currents in L_h . Moreover, it can be seen that the waveforms of the proposed design are different from those of the analysis-based design. These results indicate that the ML-based design finds a new parameter set to achieve LI operation that could never be found with the previous analysis-based designs.

Fig. 5 shows the characteristics as a function of normalized load resistance. As we can see from Fig. 5, the WPT system by the proposed design had smaller output-voltage variations against load variations than the analysis-based design. In addition, the power-delivery efficiency by the proposed design was achieved to higher power-delivery efficiency compared with the analysis-based design. It can be stated from Fig. 5 that the ML-based design can provide us with better performance of LI WPT system than the analysis-based design.

V. CONCLUSIONS

This paper has proposed the ML-based design numerical strategy for a LI WPT system. The basic concept is that the circuit design is reduced to an optimization problem solved by fully numerical computations. The WPT system, which was designed using the proposed methodology, achieved constant output voltage and higher power-delivery efficiency compared to the analysis-based design. The proposed design also enabled highly accurate performance prediction, which was verified by the quantitative agreement between experimental results and numerical predictions. The ML-based optimization finds the optimal parameter set from parameters outside the range that it had considered to be common sense, which is an unexpected but valuable result. The results of this paper show the possibility of ML-based design in power electronics circuits.

VI. ACKNOWLEDGMENT

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